De Leeuw transference on homogeneous $\dot{W}^{1,1}$

Solution packet for arXiv:1306.1437v4

12 August 2026

Status. Candidate full solution in the exact space of the source paper, likely valid, awaiting human review.

1 Source question and result

Kazaniecki and Wojciechowski prove that every Fourier multiplier on $\dot{W}^{1,1}(\mathbb{R}^d)$, $d \geq 2$, has a bounded continuous representative [1]. Their argument needs estimates on the torus. They observe that ordinary L^1 has de Leeuw transference, but write that no equivalent was known for the homogeneous Sobolev space and leave the general de Leeuw-type question open. The statement is on source PDF page 3:

> Our proof uses three main ingredients. The first one is the Bonami - Poornima result. The second is the Riesz product technique which allows us to make the crucial estimates on the torus group. This would be sufficient for our purpose, provided we are able to transfer the problem from $\mathbb{R}^d$ to $\mathbb{T}^d$. Such transference in the case of multipliers on L^p spaces is the subject of the theorem of deLeeuw (cf. [7]). However, in the case of multipliers on the homogeneous Sobolev space no version of the deLeeuw transference theorem is known. We are able to overcome this difficulty due to the special form of functions on which the multiplier reaches its norm. The question of general deLeeuw type theorem for the homogeneous Sobolev spaces remains open. We believe that this paper will provide a motivation for further research in this direction.

We prove both analogues of the classical de Leeuw theorems which the source states immediately afterward: restriction to a lattice and approximation from shrinking lattices. In particular, we obtain an exact norm identity.

Put $\mathbb{T}^d = \mathbb{R}^d/(2\pi\mathbb{Z}^d)$ and use the seminorms

$$\|f\|_{\dot{W}_{\mathbb{R}}} = \sum_{j=1}^{d} \|\partial_j f\|_{L^1(\mathbb{R}^d)}, \qquad \|P\|_{\dot{W}_{\mathbb{T}}} = \sum_{j=1}^{d} \|\partial_j P\|_{L^1(\mathbb{T}^d)}.$$

Both spaces are understood modulo constants. Multiplier norms are denoted by $\|m\|_{\mathcal{M}(\dot{W}_{\mathbb{R}})}$ and $\|\gamma\|_{\mathcal{M}(\dot{W}_{\mathbb{T}})}$. The coefficient γ_0 never affects the torus seminorm.

Proof Intuition

For restriction, test the Euclidean multiplier on a trigonometric polynomial inside a slowly varying band-limited envelope. The apparent obstruction is that a homogeneous $\dot{W}^{1,1}$ multiplier need not globally be convolution by a measure. All finitely many wave packets avoid frequency zero, however; after a smooth localization there, one can invert a coordinate derivative and turn the multiplier into an ordinary L^1 measure multiplier. Translation continuity then identifies the torus action with no loss in the limiting constant. In the reverse direction, Poisson summation turns inverse-Fourier Riemann sums into periodizations of a band-limited Schwartz function. The input norms converge on expanding period cubes, while the output gradients converge locally to the Euclidean multiplier output. The two limits give the reverse inequality and hence exact norm equality.

Theorem 1 (Homogeneous de Leeuw theorem)**.** *Let $d \geq 2$ and let $m : \mathbb{R}^d \to \mathbb{C}$ be continuous. For $\varepsilon > 0$ set*

$$\gamma_k^\varepsilon = m(\varepsilon k), \qquad k \in \mathbb{Z}^d.$$

Then the following statements hold.

1. *If $m \in \mathcal{M}(\dot{W}_\mathbb{R})$, then*

$$\gamma^\varepsilon \in \mathcal{M}(\dot{W}_\mathbb{T}), \qquad \|\gamma^\varepsilon\|_{\mathcal{M}(\dot{W}_\mathbb{T})} \leq \|m\|_{\mathcal{M}(\dot{W}_\mathbb{R})} \quad (\varepsilon > 0).$$

2. *If $\varepsilon_n \downarrow 0$ and $\sup_n \|\gamma^{\varepsilon_n}\|_{\mathcal{M}(\dot{W}_\mathbb{T})} \leq C$, then*

$$m \in \mathcal{M}(\dot{W}_\mathbb{R}), \qquad \|m\|_{\mathcal{M}(\dot{W}_\mathbb{R})} \leq C.$$

Consequently,

$$\|m\|_{\mathcal{M}(\dot{W}_\mathbb{R})} = \sup_{\varepsilon>0} \|(m(\varepsilon k))_{k\in\mathbb{Z}^d}\|_{\mathcal{M}(\dot{W}_\mathbb{T})}, \tag{1}$$

with the convention that either side may be infinite.

Because the continuity hypothesis is automatic for Euclidean multipliers by the main theorem of [1], Theorem 1 applies to every multiplier in the precise setting of that paper. It does not claim the same for every homogeneous order and exponent; see the scope note at the end.

We use the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}^d} f(x)e^{-ix\cdot\xi}\,dx, \qquad f(x) = (2\pi)^{-d}\int_{\mathbb{R}^d} \widehat{f}(\xi)e^{ix\cdot\xi}\,d\xi.$$

2 Two localization lemmas

The first observation repairs the obstacle in a direct cutoff argument: although m need not be a Fourier transform of a measure globally, every smooth localization away from the origin is one.

Lemma 2 (Local L^1 multiplier)**.** *Suppose $m \in \mathcal{M}(\dot{W}_\mathbb{R})$ and $\chi \in C_c^\infty(\mathbb{R}^d \setminus \{0\})$. Then $m\chi$ is an ordinary $L^1(\mathbb{R}^d)$ multiplier. Hence $m\chi = \widehat{\mu}$ for a finite complex Borel measure μ.*

Proof. Choose a finite smooth decomposition $\chi = \sum_r \chi_r$ such that, for each r, some coordinate ξ_{j_r} is bounded away from zero on $\operatorname{supp}\chi_r$. Given $g \in \mathcal{S}(\mathbb{R}^d)$, define

$$\widehat{f_r}(\xi) = \frac{\chi_r(\xi)}{i\xi_{j_r}}\widehat{g}(\xi).$$

The multiplier identity gives

$$T_{m\chi_r} g = \partial_{j_r} T_m f_r.$$

For every ℓ, the symbol $\xi_\ell \chi_r(\xi)/\xi_{j_r}$ is smooth and compactly supported. Its inverse Fourier transform is in L^1, so Young's inequality gives

$$\|f_r\|_{\dot W_{\mathbb{R}}} \le C_{\chi_r} \|g\|_1.$$

It follows that

$$\|T_{m\chi_r} g\|_1 \le \|T_m f_r\|_{\dot W_{\mathbb{R}}} \le \|m\|_{\mathcal{M}(\dot W_{\mathbb{R}})} C_{\chi_r} \|g\|_1.$$

Summing in r makes $m\chi$ an L^1 multiplier. The classical characterization of translation-invariant bounded operators on L^1 identifies their symbols with Fourier transforms of finite measures [2]. $\square$

For a finite set $F \subset \mathbb{Z}^d$, write $T_{b|F}(\sum_{k\in F} c_k e^{ik\cdot x}) = \sum_{k\in F} b(k) c_k e^{ik\cdot x}$.

Lemma 3 (Finite-frequency wave packets)**.** *Let $b = \widehat{\mu}$ for a finite complex measure μ, let $q(x) = \sum_{k\in F} c_k e^{ik\cdot x}$, and let $\Phi \in \mathcal{S}(\mathbb{R}^d)$. Then*

$$\delta^d \big\| T_b(\Phi(\delta\,\cdot) q) - \Phi(\delta\,\cdot) T_{b|F} q \big\|_1 \longrightarrow 0 \qquad (\delta \downarrow 0). \tag{2}$$

Proof. For one frequency, convolution by μ and the change of variables $z = \delta x$ bound the normalized error by

$$\int_{\mathbb{R}^d} \|\Phi(\,\cdot - \delta y) - \Phi\|_1 \, d|\mu|(y).$$

This tends to zero by L^1 translation continuity and dominated convergence, since the integrand is at most $2\|\Phi\|_1$. Sum over the finite set F. $\square$

We also use the elementary periodic averaging formula. If a is 2π-periodic and integrable on $\mathbb{T}^d$, while $\Phi \in \mathcal{S}$ is nonnegative, then

$$\delta^d \int_{\mathbb{R}^d} \Phi(\delta x) |a(x)| \, dx \longrightarrow A_\Phi \int_{\mathbb{T}^d} |a(y)| \, dy, \qquad A_\Phi = (2\pi)^{-d} \int_{\mathbb{R}^d} \Phi. \tag{3}$$

One proves this first for continuous a by dividing $\mathbb{R}^d$ into period cubes and taking a Riemann sum, then obtains the stated form by L^1 approximation.

3 Restriction to the torus

It suffices first to sample on $\mathbb{Z}^d$. Let P be a trigonometric polynomial with zero constant term and finite spectrum $F \subset \mathbb{Z}^d \setminus \{0\}$. Choose a nonzero Schwartz function ψ whose Fourier transform has compact support and put $\Phi = |\psi|^2$. Thus $\Phi \ge 0$, $\Phi \in \mathcal{S}$, and $\widehat{\Phi}$ is compactly supported. Set

$$f_\delta(x) = \Phi(\delta x) P(x).$$

For all sufficiently small δ, the Fourier support of f_δ is contained in a fixed compact subset of $\mathbb{R}^d \setminus \{0\}$. Choose $\chi \in C_c^\infty(\mathbb{R}^d \setminus \{0\})$ equal to one on that set and put $b = m\chi$. Lemma 2 applies, and $T_m f_\delta = T_b f_\delta$.

The product rule, (3), and the extra factor δ in the derivative of the envelope give

$$\delta^d \|f_\delta\|_{\dot W_{\mathbb{R}}} \longrightarrow A_\Phi \|P\|_{\dot W_{\mathbb{T}}}. \tag{4}$$

Indeed,

$$\partial_j f_\delta = \Phi(\delta x)\partial_j P(x) + \delta(\partial_j\Phi)(\delta x)P(x),$$

and the normalized L^1 norm of the second term is $O(\delta)$.

Let $\gamma_k = m(k)$. Commutation with derivatives gives

$$\partial_j T_m f_\delta = T_b\big(\Phi(\delta\,\cdot)\partial_j P\big) + \delta T_b\big((\partial_j\Phi)(\delta\,\cdot)P\big).$$

The last term again has normalized norm $O(\delta)$ because convolution by the finite measure representing b is L^1 bounded. Lemma 3 applied to $\partial_j P$, followed by (3), yields

$$\delta^d\|T_m f_\delta\|_{\dot W_{\mathbb{R}}} \longrightarrow A_\Phi\|T_\gamma P\|_{\dot W_{\mathbb{T}}}. \tag{5}$$

Combining (4), (5), and the Euclidean multiplier inequality proves

$$\|T_\gamma P\|_{\dot W_{\mathbb{T}}} \le \|m\|_{\mathcal{M}(\dot W_{\mathbb{R}})}\|P\|_{\dot W_{\mathbb{T}}}.$$

Trigonometric polynomials modulo constants are dense, so γ extends to the asserted torus multiplier. Finally, the symbol $m_\varepsilon(\xi) = m(\varepsilon\xi)$ has the same Euclidean multiplier norm as m by dilation. Applying the unit-lattice result to m_ε proves part 1 of Theorem 1.

4 Approximation from shrinking lattices

Assume $\varepsilon_n \downarrow 0$ and $\|\gamma^{\varepsilon_n}\|_{\mathcal{M}(\dot W_{\mathbb{T}})} \le C$. Testing on the nonconstant exponentials shows that $|m(\varepsilon_n k)| \le C$ for every $k \ne 0$. Given $\xi \in \mathbb{R}^d$, choose nonzero $k_n \in \mathbb{Z}^d$ with $\varepsilon_n k_n \to \xi$; continuity then gives $|m(\xi)| \le C$. Thus m is bounded, as required in the multiplier definition. We first take $f \in \mathcal{S}(\mathbb{R}^d)$ with $\widehat{f} \in C_c^\infty(\mathbb{R}^d)$. Put $L_n = 2\pi/\varepsilon_n$ and define the finite trigonometric polynomial

$$P_n(y) = \sum_{k\in\mathbb{Z}^d}\left(\frac{\varepsilon_n}{2\pi}\right)^d \widehat{f}(\varepsilon_n k)e^{ik\cdot y}. \tag{6}$$

Poisson summation gives, on the central cube Q_{L_n} of side L_n,

$$P_n(\varepsilon_n x) = F_n(x) := \sum_{\ell\in\mathbb{Z}^d} f(x + L_n\ell). \tag{7}$$

For each derivative, the period cubes partition $\mathbb{R}^d$. Hence the triangle inequality gives the upper bound by the total Euclidean norm, while the reverse triangle inequality on the central summand loses only the exterior tail. Therefore

$$\sum_{j=1}^d \int_{Q_{L_n}} |\partial_j F_n(x)|\,dx \longrightarrow \|f\|_{\dot W_{\mathbb{R}}}. \tag{8}$$

The rescaled output derivatives are

$$\begin{aligned} H_{n,j}(x) &:= \varepsilon_n(\partial_{y_j} T_{\gamma^{\varepsilon_n}} P_n)(\varepsilon_n x) \\ &= \sum_{k\in\mathbb{Z}^d}\left(\frac{\varepsilon_n}{2\pi}\right)^d i\varepsilon_n k_j m(\varepsilon_n k)\widehat{f}(\varepsilon_n k)e^{i\varepsilon_n k\cdot x}. \end{aligned} \tag{9}$$

For every fixed compact set of x's, this is a Riemann sum converging uniformly to

$$G_j(x) = (2\pi)^{-d} \int_{\mathbb{R}^d} i\xi_j m(\xi)\widehat{f}(\xi)e^{ix\cdot\xi}\, d\xi = \partial_j T_m f(x). \tag{10}$$

Indeed, the integrand is continuous and compactly supported, and its modulus of continuity in ξ is uniform for x in a fixed compact set.

Changing variables $y = \varepsilon_n x$ in the torus multiplier inequality cancels the common factor ε_n^{1-d} and gives

$$\sum_j \int_{Q_{L_n}} |H_{n,j}(x)|\, dx \le C \sum_j \int_{Q_{L_n}} |\partial_j F_n(x)|\, dx. \tag{11}$$

Fix a ball B_R. For large n, it lies in Q_{L_n}. Uniform convergence in (9), (8), and (11) show that

$$\sum_j \int_{B_R} |G_j(x)|\, dx \le C\|f\|_{\dot{W}_{\mathbb{R}}}.$$

Letting $R \to \infty$ proves

$$\|T_m f\|_{\dot{W}_{\mathbb{R}}} \le C\|f\|_{\dot{W}_{\mathbb{R}}} \tag{12}$$

on the band-limited Schwartz class.

For completeness, that class is dense in $\dot{W}^{1,1}(\mathbb{R}^d)$ modulo constants when $d \ge 2$. The endpoint Sobolev inequality supplies a representative $u \in L^{d/(d-1)}$ for every homogeneous class. If $\eta_R(x) = \eta(x/R)$ is a standard cutoff, then

$$\|(1-\eta_R)\nabla u\|_1 \to 0, \qquad \|u\nabla\eta_R\|_1 \lesssim \|u\|_{L^{d/(d-1)}(\{R \lesssim |x| \lesssim 2R\})} \to 0.$$

Mollification therefore proves density of C_c^∞. Convolving a compactly supported smooth function with an approximate identity $\rho_t \in \mathcal{S}$ for which $\widehat{\rho}_t$ is compactly supported gives band-limited Schwartz approximants and convergence of every first derivative in L^1. Thus (12) extends uniquely to all of $\dot{W}_{\mathbb{R}}$, proving part 2. Combining the two parts proves (1).

5 Scope and novelty note

The theorem is complete for the endpoint first-order homogeneous space studied in [1]: restriction, shrinking-lattice approximation, and exact norm equality. It does not settle all orders and exponents; the low-degree polynomial quotients and test-class density across homogeneous Sobolev thresholds require separate analysis.

A literature screen through 12 August 2026 found later continuity results on $\dot{W}^{l,1}$ and $\dot{W}^{l,\infty}$ [3], and related transference for the *inhomogeneous* $W^{l,1}$ spaces [4]. Neither source states the homogeneous shrinking-lattice theorem above. This is a bounded, not exhaustive, novelty screen.

References

[1] K. Kazaniecki and M. Wojciechowski, *On the continuity of Fourier multipliers on the homogeneous Sobolev spaces* $\dot{W}_1^1(\mathbb{R}^d)$, Ann. Inst. Fourier 66 (2016), no. 3, 1247–1260; arXiv:1306.1437.

[2] W. Rudin, *Fourier Analysis on Groups*, Wiley, 1962.

[3] E. Curca, *On the continuity of Fourier multipliers on* $\dot{W}^{l,1}(\mathbb{R}^d)$ *and* $\dot{W}^{l,\infty}(\mathbb{R}^d)$, J. Funct. Anal. (2022), 109573.

[4] E. Curca and M. Wojciechowski, *Nonexistence of Henkin type projections via a Wiener theorem for multipliers*, arXiv:2604.23161 (2026).